Measurement, Observables, and the Quantum Toolbox

EVA: Quantum Machine Learning – Lecture 3

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Roadmap

1. Projective measurement and shot statistics
2. Observables and expectation values
3. Qiskit primitives: sampler vs estimator
4. Complex numbers in quantum and ML
5. Real-data mini experiment (Iris, two classes)

Measurement and shots

Projective measurement in the computational basis

For a single qubit

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,, \qquad |\alpha|^2 + |\beta|^2 = 1,$$

measurement in the Z basis returns:

$$P(0) = |\alpha|^2, \qquad P(1) = |\beta|^2.$$

After measurement, the state collapses to the observed basis state. A single shot gives one random bit; many shots estimate probabilities.

Why we need multiple shots

Let $p = P(1)$ and $\hat{p} = k/N$ from N shots.

$$\mathbb{E}[\hat{p}] = p, \qquad \mathrm{Var}(\hat{p}) = \frac{p(1-p)}{N}.$$

Typical standard error:

$$\sigma_{\hat{p}} \approx \sqrt{\frac{p(1-p)}{N}}$$

scales as $1/\sqrt{N}$.

`lecture_03_demo.ipynb`, Demo 1: shot convergence.

Hoeffding bound for shot planning

For bounded outcomes in $[0, 1]$:

$$\Pr\left(|\hat{p} - p| \geq \epsilon\right) \leq 2e^{-2N\epsilon^2}.$$

To guarantee error ϵ with confidence $1 - \delta$:

$$N \geq \frac{\ln(2/\delta)}{2\epsilon^2}.$$

Example: $\epsilon = 0.01$, $\delta = 0.05$

$$N \gtrsim 18{,}445.$$

This is a worst-case bound; practical circuits may need fewer shots.

Observables and expectation values

Observables

In quantum mechanics, measurable quantities are Hermitian operators.

Single-qubit Pauli observables:

$$I,\ X,\ Y,\ Z.$$

Multi-qubit examples:

$$Z \otimes Z, \quad X \otimes X, \quad Z \otimes I.$$

Measurement outcomes are eigenvalues (for Pauli operators: ± 1).

Expectation values

For state $|\psi\rangle$ and observable O:

$$\langle O \rangle = \langle \psi | \, O \, | \psi \rangle \, .$$

Interpretation:

- one shot gives one random eigenvalue
- many shots estimate the mean $\langle O \rangle$
- this mean is the quantity used in variational QML loss functions

Bell-state correlations

For

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle),$$

key expectation values:

$$\langle Z \otimes Z\rangle = +1, \qquad \langle X \otimes X\rangle = +1, \qquad \langle Z \otimes I\rangle = 0.$$

Reading:

- perfect joint correlation in Z and X
- each local marginal is unbiased

`lecture_03_demo.ipynb`, Demo 2: Bell observables.

Sampler vs estimator

Qiskit primitives: conceptual split

Primitive	Input	Output
Sampler	measured circuit	bitstring distribution
Estimator	circuit + observable	expectation value

Rule of thumb:

- use sampler for histograms and counts
- use estimator for variational objectives and observables

Same circuit, two questions

```
# Sampler question: what outcomes occur?
qc = QuantumCircuit(1)
qc.ry(theta, 0)
qc.measure_all()

# Estimator question: what is <Z>?
qc_no_meas = QuantumCircuit(1)
qc_no_meas.ry(theta, 0)
```

`lecture_03_demo.ipynb`, Demo 3: sampler/estimator comparison.

Complex numbers in quantum and ML

Why complex numbers are operational, not decorative

Complex amplitudes carry phase information that controls interference.

Two states can share the same Z-basis probabilities but differ in phase, yielding different values of $\langle X \rangle$ or $\langle Y \rangle$.

This is why phase-sensitive observables matter in quantum modeling.

Connection to machine learning

Classical models usually use real-valued features and weights.

Quantum states are naturally complex-valued feature vectors in Hilbert space.

Practical bridge used in this course:

- map real data into quantum states
- extract expectation-value features (e.g., $\langle X \rangle, \langle Y \rangle, \langle Z \rangle$)
- train a classical model on these features

Real-data mini experiment (Iris)

In this lecture, we use a two-class Iris subset and compare:

1. baseline logistic regression on raw standardized features
2. quantum-feature baseline from angle encoding with observables $\langle X \rangle, \langle Y \rangle, \langle Z \rangle$

Goal: not advantage claims, but pipeline literacy from real data to quantum-style feature extraction.

`lecture_03_demo.ipynb`, Demo 4: Iris complex/quantum features.

Summary

- measurement is probabilistic, expectation values are statistical means
- shot count controls estimation precision
- observables define what quantity is being measured
- sampler and estimator answer different questions
- real datasets can already be integrated at the measurement stage

Next class: classical ML refresher and kernel methods, then quantum feature maps.

References i

1. Nielsen and Chuang, *Quantum Computation and Quantum Information*.
2. Schuld and Petruccione, *Supervised Learning with Quantum Computers*.
3. Qiskit documentation: `https://docs.quantum.ibm.com/`.
4. PennyLane documentation: `https://docs.pennylane.ai/`.